

Most component mechanisms are already owned by mature parent fields

Native ownership (rows) of scientific decisions (columns). The proposed field claims none of these columns; only the cross-row composition gaps (right) are under test.

■ native core ownership

▨ partial / contributing ownership

◇ candidate composition gap (a parent composition may suffice)

□ blank cell: not a native object of that field

		C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11
1	Formal learning & computational epistemology [10]	■			▨							
2	Cybernetics, control & resilience engineering [18-20]		▨		▨							
3	Metareasoning & active learning [15-17] †		■									
4	Belief revision, truth maintenance & diagnosis [11-14] ‡			■	■							
5	Formal abstraction & behavioural equivalence [24, 25]					■						
6	Causal transportability & measurement invariance [21, 22]					■	■					
7	Metrology & traceability [23]						■					
8	Statistical workflow & calibration [27-29] ‡				▨				▨	■		
9	Software testing & oracles [32, 33] †				■					■		
10	Numerical analysis & validated computation [30, 31]				▨	▨						
11	Provenance & reproducibility [35, 36, 54]							■				
12	Dependent evidence synthesis [34, 37]								■	■		
13	Assurance, argumentation & consensus [38-41] †								■	▨	▨	
14	Social epistemology & philosophy of inquiry [42-50]			▨					▨		■	
15	Sociology of science & indigenous data governance [51-53, 7]						▨				■	
16	Systematic review, metascience & performative evaluation [65, 66, 69]									■		▨
17	AI for Science & autonomous laboratories [1-5]		▨					▨				■
18	Epistemic control & spec-driven architectures [59-61]							▨		▨	■	

CANDIDATE COMPOSITION GAPS

each exists only where multiple rows must interact

◇ provenance -> scientific validity

needs rows 9, 11, 12, 13

◇ relation -> downstream reuse / reopening

needs rows 4, 5, 6

◇ evidence -> authority-bounded adoption

needs rows 12, 14, 15

◇ route stop -> scientific closure

needs rows 4, 16, 18

◇ local repair -> search-space escalation

needs rows 2, 3, 4

a parent composition may suffice:

if it makes the same protected decision, the field hypothesis contracts

C1 reliable learning & identifiability C2 action & experiment selection C3 belief dependency & revision C4 diagnosis & discrimination

C5 relation & transport C6 measurement & linking C7 provenance & replay C8 evidence combination

C9 evaluator response C10 authority & permission C11 end-to-end automation

† bibliographic correction applied to this row's canonical source during the 2026-08-29 citation audit (one detail still unpinpointed for ref 38); see manuscript reference list.

‡ minor bibliographic fix applied (refs 12, 29).

Row sources are the manuscript's audited canonical references; no row cites an uncorrected wrong-attributed source.